

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} 0 & 0 & G_z \end{bmatrix}$$

$$G_z = G_{pz}^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} G_{xx} & G_{xy} & 0 \\ G_{xy} & -G_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xx} = G_{dv}^{(1)}$$

$$G_{xy} = G_{dxy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & G_{xxz} \\ 0 & 0 & G_{xxz} \\ 0 & 0 & G_{zzz} \\ 0 & G_{xxz} & 0 \\ G_{xxz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxz} = -G_{faz}^{(1)} + G_{pz}^{(1)}$$

$$G_{zzz} = 2G_{faz}^{(1)} + 3G_{pz}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & G_{xxz} \\ 0 & 0 & G_{xxz} \\ 0 & 0 & 0 \\ G_{yzx} & -\frac{G_{xxz}}{2} & 0 \\ -\frac{G_{xxz}}{2} & -G_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxz} = 2G_{pz}^{(2)}$$

$$G_{yzx} = 3Q_{du}^{(1)}$$

$$\frac{G_{xxz}}{2} = G_{pz}^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & G_{yzy} & 0 \\ -G_{yzy} & G_{yzx} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = -Q_{du}^{(2)} + Q_s^{(1)}$$

$$G_{yzy} = G_{pz}^{(3)}$$

$$G_{xyz} = 2Q_{du}^{(2)} + Q_s^{(1)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} G_{xxxx} & 0 & G_{xxzz} & 0 & 0 & G_{xxxxy} \\ 0 & -G_{xxxx} & -G_{xxzz} & 0 & 0 & G_{xxxxy} \\ G_{xxzz} & -G_{xxzz} & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & G_{yzyz} & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{yzzx} & -G_{yzyz} & 0 \\ G_{xxxxy} & G_{xxxxy} & G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxxx} = 2G_{dv}^{(1)} + 4G_{dv}^{(2)} + G_{gv}^{(1)}$$

$$G_{xxzz} = G_{dv}^{(1)} - G_{gv}^{(1)}$$

$$G_{xxyy} = G_{dxy}^{(1)} + 2G_{dxy}^{(2)} - G_{gbz}^{(1)}$$

$$G_{zzxy} = G_{dxy}^{(1)} + 2G_{gbz}^{(1)}$$

$$G_{yyzz} = -G_{dv}^{(2)} + G_{gv}^{(1)}$$

$$G_{yzzx} = G_{dxy}^{(2)} + 2G_{gbz}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & G_{xxyy} & G_{xxzz} & 0 & 0 & G_{xxyy} \\ -G_{xxyy} & 0 & -G_{xxzz} & 0 & 0 & G_{xxyy} \\ -G_{xxzz} & G_{xxzz} & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -G_{xxyy} & -G_{xxyy} & -G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$G_{xxyy} = -2G_{dv}^{(3)} + 4Q_{f3}^{(1)}$$

$$G_{xxzz} = -G_{dv}^{(3)} - 4Q_{f3}^{(1)}$$

$$G_{xxyy} = G_{dxy}^{(3)} + 2Q_{fbz}^{(1)}$$

$$G_{zzxy} = G_{dxy}^{(3)} - 4Q_{fbz}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} G_{yyzz} & G_{yzzx} & 0 \\ G_{yzzx} & -G_{yyzz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{yyzz} = 2G_{dv}^{(4)}$$

$$G_{yzzx} = 2G_{dxy}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & G_{xxyy} \\ 0 & 0 & -G_{xxyy} \\ 0 & 0 & 0 \\ G_{yyzz} & G_{yzzx} & 0 \\ -G_{yzzx} & G_{yyzz} & 0 \\ 0 & 0 & G_{xyxy} \end{bmatrix}$$

$$G_{xxyy} = -2G_{dxy}^{(5)} + Q_{fbz}^{(2)}$$

$$G_{yyzz} = -G_{dv}^{(5)} + Q_{f3}^{(2)}$$

$$G_{yzzx} = -G_{dxy}^{(5)} - Q_{fbz}^{(2)}$$

$$G_{xyxy} = 2G_{dv}^{(5)} + Q_{f3}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & G_{yyyz} & G_{yzzx} & 0 \\ 0 & 0 & 0 & -G_{yzzx} & G_{yyyz} & 0 \\ G_{xyxx} & -G_{xyxx} & 0 & 0 & 0 & G_{xyxy} \end{bmatrix}$$

$$G_{yyyz} = -G_{dv}^{(6)} + Q_{f3}^{(3)}$$

$$G_{yzzx} = G_{dxy}^{(6)} + Q_{fbz}^{(3)}$$

$$G_{xyxx} = -2G_{dxy}^{(6)} + Q_{fbz}^{(3)}$$

$$G_{xyxy} = 2G_{dv}^{(6)} + Q_{f3}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} G_{xxxx} & G_{xxyy} & G_{xxzz} & 0 & 0 & G_{xxyy} & 0 & 0 & G_{xxyx} \\ -G_{xxyy} & -G_{xxxx} & -G_{xxzz} & 0 & 0 & G_{xxyx} & 0 & 0 & G_{xxxy} \\ G_{zzxx} & -G_{zzxx} & 0 & 0 & 0 & G_{zzxy} & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & G_{yzyz} & G_{yzxz} & 0 & G_{yzzy} & G_{yzxz} & 0 \\ 0 & 0 & 0 & G_{yzxz} & -G_{yzzy} & 0 & G_{yzxz} & -G_{yzyz} & 0 \\ G_{xyxx} & G_{xyxx} & G_{xyzx} & 0 & 0 & G_{xyxy} & 0 & 0 & -G_{xyxy} \end{bmatrix}$$

$$\begin{aligned} G_{xxxx} &= 2G_{dv}^{(1)} + 4G_{dv}^{(2)} + G_{gv}^{(1)} \\ G_{xxyy} &= -2G_{dv}^{(3)} + 4Q_{f3}^{(1)} \\ G_{xxzz} &= G_{dv}^{(1)} - G_{dv}^{(3)} - G_{gv}^{(1)} - 4Q_{f3}^{(1)} \\ G_{xxyy} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} - 2G_{dxy}^{(5)} - G_{gbz}^{(1)} + 2Q_{fbz}^{(1)} + Q_{fbz}^{(2)} \\ G_{xxyx} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} + 2G_{dxy}^{(5)} - G_{gbz}^{(1)} + 2Q_{fbz}^{(1)} - Q_{fbz}^{(2)} \\ G_{zzxx} &= G_{dv}^{(1)} + G_{dv}^{(3)} - G_{gv}^{(1)} + 4Q_{f3}^{(1)} \\ G_{zzxy} &= G_{dxy}^{(1)} + G_{dxy}^{(3)} + 2G_{gbz}^{(1)} - 4Q_{fbz}^{(1)} \\ G_{yzyz} &= -G_{dv}^{(2)} - G_{dv}^{(5)} + G_{gv}^{(1)} + Q_{f3}^{(2)} \\ G_{yzxz} &= G_{dxy}^{(2)} - G_{dxy}^{(5)} + 2G_{gbz}^{(1)} - Q_{fbz}^{(2)} \\ G_{yzzy} &= -G_{dv}^{(2)} + G_{dv}^{(5)} + G_{gv}^{(1)} - Q_{f3}^{(2)} \\ G_{yzxz} &= G_{dxy}^{(2)} + G_{dxy}^{(5)} + 2G_{gbz}^{(1)} + Q_{fbz}^{(2)} \\ G_{xyxx} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} - G_{dxy}^{(3)} - G_{gbz}^{(1)} - 2Q_{fbz}^{(1)} \\ G_{xyzx} &= G_{dxy}^{(1)} - G_{dxy}^{(3)} + 2G_{gbz}^{(1)} + 4Q_{fbz}^{(1)} \\ G_{xyxy} &= 2G_{dv}^{(5)} + Q_{f3}^{(2)} \end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & G_{yzyz} & G_{yzxz} & 0 & G_{yzzy} & G_{yzxz} & 0 \\ 0 & 0 & 0 & -G_{yzxz} & G_{yzzy} & 0 & -G_{yzxz} & G_{yzzy} & 0 \\ G_{xyxx} & -G_{xyxx} & 0 & 0 & 0 & G_{xyxy} & 0 & 0 & G_{xyxy} \end{bmatrix}$$

$$\begin{aligned} G_{yzyz} &= 2G_{dv}^{(4)} - G_{dv}^{(6)} + Q_{f3}^{(3)} \\ G_{yzxz} &= 2G_{dxy}^{(4)} + G_{dxy}^{(6)} + Q_{fbz}^{(3)} \\ G_{yzzy} &= -2G_{dv}^{(4)} - G_{dv}^{(6)} + Q_{f3}^{(3)} \\ G_{yzxz} &= -2G_{dxy}^{(4)} + G_{dxy}^{(6)} + Q_{fbz}^{(3)} \\ G_{xyxx} &= -2G_{dxy}^{(6)} + Q_{fbz}^{(3)} \\ G_{xyxy} &= 2G_{dv}^{(6)} + Q_{f3}^{(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} G_{xxxx} & G_{xxxx} & 0 \\ G_{xxxx} & -G_{xxxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & G_{yyzz} \\ G_{zzxx} & G_{zzxy} & 0 \\ G_{xxyx} & -G_{yyzz} - G_{zzxx} & 0 \\ G_{yzxz} & G_{yzzy} & 0 \\ 0 & 0 & G_{zzxz} \\ -G_{yzzy} - G_{zzxz} & G_{xyyy} & 0 \\ 0 & 0 & -\frac{G_{xxxx}}{2} - \frac{G_{xxyy}}{2} + G_{xyyy} + G_{zzxy} \end{bmatrix}$$

$$\begin{aligned}
G_{xxxx} &= 2G_{dv}^{(1)} + 4G_{dv}^{(2)} + G_{gv}^{(1)} \\
G_{xxxxy} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} - 2G_{dxy}^{(5)} - G_{gbz}^{(1)} \\
G_{yyzz} &= -G_{dv}^{(1)} + G_{dv}^{(3)} + G_{gv}^{(1)} \\
G_{zzxx} &= G_{dv}^{(1)} + G_{dv}^{(3)} - G_{gv}^{(1)} \\
G_{zzxy} &= G_{dxy}^{(1)} + G_{dxy}^{(3)} + 2G_{gbz}^{(1)} \\
G_{xxyx} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} + 2G_{dxy}^{(5)} - G_{gbz}^{(1)} \\
G_{yzzx} &= G_{dxy}^{(2)} - G_{dxy}^{(5)} + 2G_{gbz}^{(1)} \\
G_{yzzz} &= -G_{dv}^{(2)} + G_{dv}^{(5)} + G_{gv}^{(1)} \\
G_{zzxz} &= G_{dv}^{(2)} + G_{dv}^{(5)} - G_{gv}^{(1)} \\
G_{xyyy} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} - G_{dxy}^{(3)} - G_{gbz}^{(1)}
\end{aligned}$$